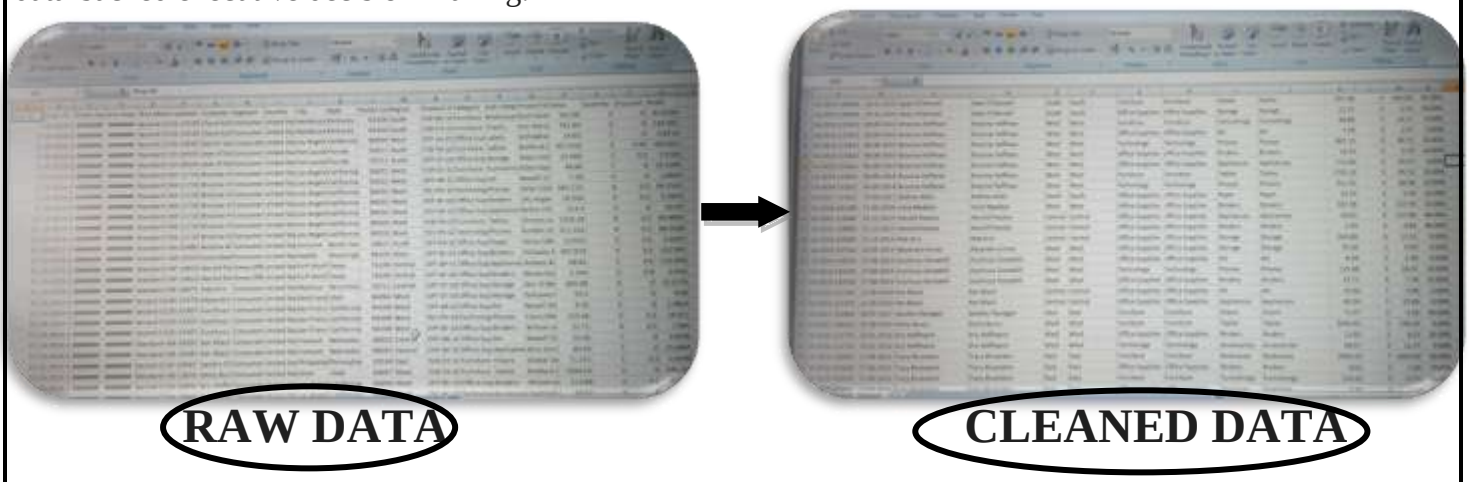
Business Analysis Report: Superstore Performance Analysis

1. Project Introduction

This business analytics project delivers an end-to-end evaluation of the **Sample Superstore** retail dataset. The objective is to analyze transaction patterns across regions, customer segments, product categories, and order timelines. By combining **Excel data wrangling**, **SQLite database extraction**, and an interactive **Power BI dashboard**, this analysis identifies key revenue engines, isolated profit leaks, and margin drivers to support data-backed executive decision-making.



2. Data Cleaning Process

Before loading the data into SQLite and Power BI, a structured data preparation process was executed in Microsoft Excel:

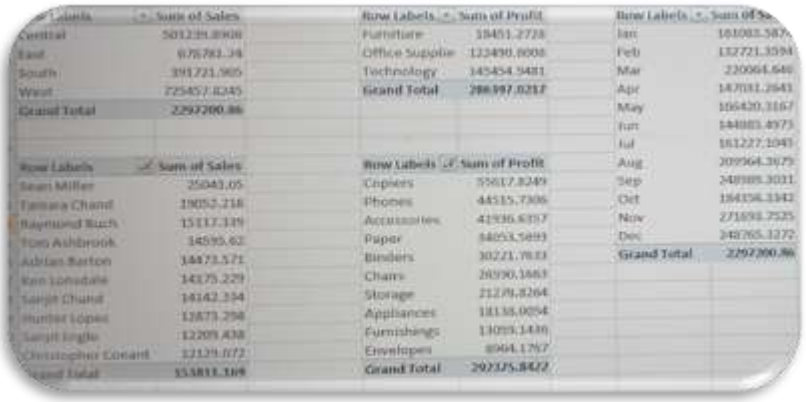
- **Duplicate Removal:** Checked the full transactional records for duplicate rows and ensured primary transaction integrity across all order lines.
- **Missing Value Handling:** Verified null and blank entries across critical fields including `Sales`, `Profit`, `Discount`, `Quantity`, and geographical dimensions.
- **Standardization:** Normalized categorical fields (such as consistent casing in `Region`, `Category`, and `Segment`) and stripped unwanted spaces.
- **Data Type Integrity:** Corrected `Order Date` and `Ship Date` into standard date formats and cast monetary measures (`Sales`, `Profit`) to numerical decimals, ensuring seamless schema creation and accurate aggregation across downstream tools.

3. Total Sales and Profit Analysis

The enterprise generated robust top-line revenue accompanied by a healthy net profit margin:

- **Total Sales: $2,297,200.86**
- **Total Profit: $286,397.02**
- **Overall Profit Margin: ~12.47%**
- **Total Distinct Orders: 5,009 transactions** across Consumer, Corporate, and Home Office segments.

While overall revenue generation is strong, the conversion rate from sales to bottom-line profit varies sharply across sub-categories and regions, primarily influenced by pricing strategy and discount elasticity.



PIVOT TABLE

4. Highest Sales Region

The **West Region** emerged as the top-performing territory across both revenue volume and profit generation:

Region	Total Sales ($)	Total Profit ($)	Profit Margin (%)	Performance Status
West	**$725,457.93**	**$108,418.79**	**14.94%**	**Leading Growth Driver**
East	$678,781.36	$91,522.84	13.48%	High Revenue & Margin
Central	$501,239.88	$39,706.45	7.92%	Margin-Diluted
South	$391,721.90	$46,749.71	11.93%	Consistent Mid-Tier

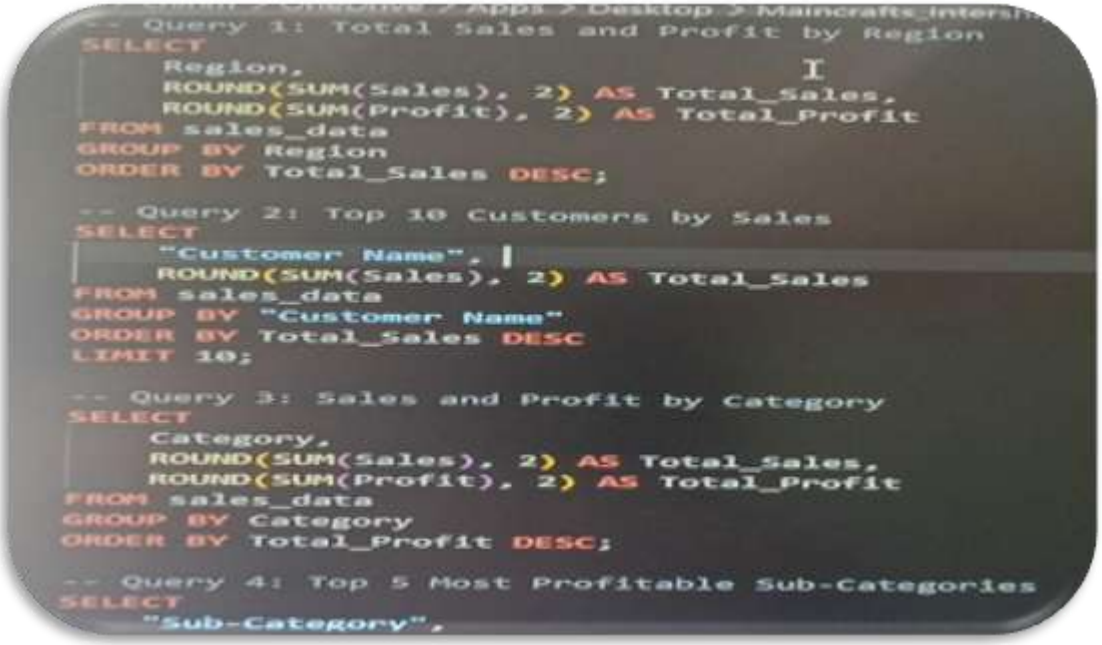
The West and East regions collectively account for nearly **70% of total company profits**. In contrast, the Central region suffered a steep drop in margin efficiency due to aggressive local promotional discounting.

5. Most Profitable Category

Among the three product categories, **Technology** delivered the highest dollar return and profit margin:

Category	Total Sales ($)	Total Profit ($)	Margin Rank	Primary Driver
Technology	**$836,154.03**	**$145,454.95**	**1st (~17.4%)**	Copiers, Phones, Accessories
Office Supplies	$719,047.03	$122,490.80	2nd (~17.0%)	Paper, Storage, Binders
Furniture	$741,999.80	$18,451.27	3rd (~2.5%)	Depressed by Tables & Bookcases

- **Technology** maintained consistent profitability with sub-categories like **Copiers** and **Phones** driving high transaction values.
- **Furniture** represents a major operational drag: despite pulling in over $740K in sales, it captured only **$18,451.27** in profit, crippled by substantial losses in the **Tables** sub-category.



```sql
-- Query 1: Total Sales and Profit by Region
SELECT
    Region,
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit
FROM sales_data
GROUP BY Region
ORDER BY Total_Sales DESC;

-- Query 2: Top 10 Customers by Sales
SELECT
    "Customer Name",
    ROUND(SUM(Sales), 2) AS Total_Sales
FROM sales_data
GROUP BY "Customer Name"
ORDER BY Total_Sales DESC
LIMIT 10;

-- Query 3: Sales and Profit by Category
SELECT
    Category,
    ROUND(SUM(Sales), 2) AS Total_Sales,
    ROUND(SUM(Profit), 2) AS Total_Profit
FROM sales_data
GROUP BY Category
ORDER BY Total_Profit DESC;

-- Query 4: Top 5 Most Profitable Sub-Categories
SELECT
    "Sub-Category",
```

SQL QUERY

6. Discount Impact

Discount strategy is the single largest factor separating profitable transactions from loss-making orders:

- **The Negative Threshold (>20%):** Products sold at a discount rate exceeding **20%** consistently produce net negative margins.
- **Volume Fallacy:** High discounts stimulated order volume but failed to generate incremental operating margin. Instead, they led to negative gross margin realization after freight and handling overheads.
- **The Worst Hit Sub-Category: Tables** consistently carried average discounts between 30% and 40%, directly explaining why the sub-category accumulated thousands of dollars in cumulative net losses.

7. Best Performing Month

Temporal analysis from the Power BI Line Chart reveals marked fourth-quarter seasonality:

- **Peak Revenue Month: November** stands out as the single highest-performing month by revenue and order count, followed closely by **December**.
- **Seasonal Surge:** Over **30% of annual revenue** is concentrated in Q4 (September to December), driven by corporate fiscal budget-clearing cycles and holiday retail demand.
- **Low Period: January and February** reflect recurring post-holiday demand troughs with the lowest monthly transaction volumes of the calendar year.

8. Management Recommendations

1. **Enforce Hard Discount Caps:** Establish an automatic discount ceiling of **15% on Furniture lines** (especially Tables and Bookcases). Any discount beyond 20% should mandate managerial sign-off.
2. **Prioritize Technology & Office Supplies:** Reallocate promotional and ad-spend budgets toward high-margin lines (**Copiers, Accessories, and Paper**) where gross profit conversion exceeds 17%.
3. **Address Central Region Profit Drag:** Overhaul the promotional strategy in the Central region to mirror the pricing discipline of the West and East regions.
4. **Q4 Demand & Inventory Planning:** Align supply chain and warehouse stocking ahead of the August–September period to ensure zero stockouts during the peak November–December cycle.
5. **Key Account Loyalty Program:** Develop targeted tier-based incentives for top enterprise customers (such as Sean Miller and Tamara Chand) to protect high-volume, low-churn recurring revenue.



Sales_Analysis_Dashboard

9. Conclusion

The Superstore business maintains strong revenue momentum led by the West region and the Technology category. However, overall enterprise profitability is actively eroded by uncontrolled discounting in Furniture (specifically Tables) and compressed margins in the Central region. Implementing strict discount controls, prioritizing high-return inventory, and capitalizing on Q4 seasonal peaks will substantially elevate operating margins without compromising total sales volume.